Name bhavya rana
Sapid 590034689
Batch 15

Algorithm

Main Program

1. Start.
2. Display the menu:
    - Reverse each word in a sentence.
    - Print initials of a name.
    - Exit.
3. Accept the user's choice.
4. Use `switch` to call the corresponding function.
5. Repeat until the user selects Exit.

Function 1: Reverse each word

1. Read a sentence.
2. Scan the sentence character by character.
3. Identify the beginning and end of each word.
4. Reverse only the characters within that word.
5. Keep spaces and the order of words unchanged.
6. Display the modified sentence.

Example:
`What is special?` → `tahW si ?laiceps`

Note: Reversing characters preserves their original case, so `What` becomes `tahW`. If the required output must specifically be `tahw`, the program would also need to convert the uppercase `W` to lowercase.

Function 2: Print initials

1. Read the full name.
2. Print the first character as an initial.
3. Scan the name for spaces.
4. Whenever a space is followed by a character, print that character as the next initial.
5. Add `.` after each initial.

Example:
`Alexander Grahambell` → `A.G.`

Test Case Table

Choice	Input	Expected Output
1	What is special?	tahW si ? laiceps
1	Hello World	olleH dlroW
1	C Programming	C gnimmargorP
2	Alexander Grahambell	A.G.
2	Bhavya Rana	B.R.
2	John Michael Smith	J.M.S.
3	—	Program terminates
5	—	Invalid choice!

C Program

```
#include <stdio.h>
#include <string.h>

void reverseEachWord(void);
void printInitials(void);

int main()
{
    int choice;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();  // Remove newline left by scanf()

        switch (choice)
        {
            case 1:
                reverseEachWord();
                break;

            case 2:
```

```
                printInitials();
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 3);

    return 0;
}

/* Function to reverse each word without changing word order
*/
void reverseEachWord(void)
{
    char sentence[200];
    int i, start, end, len;
    char temp;

    printf("Enter a sentence: ");
    fgets(sentence, sizeof(sentence), stdin);

    sentence[strcspn(sentence, "\n")] = '\0';

    len = strlen(sentence);
    start = 0;

    for (i = 0; i <= len; i++)
    {
        if (sentence[i] == ' ' || sentence[i] == '\0')
        {
            end = i - 1;

            while (start < end)
            {
                temp = sentence[start];
                sentence[start] = sentence[end];
                sentence[end] = temp;
```

```
                start++;
                end--;
            }

            start = i + 1;
        }
    }

    printf("Sentence after reversing each word: %s\n",
sentence);
}

/* Function to print initials of a name */
void printInitials(void)
{
    char name[200];
    int i, len;

    printf("Enter the full name: ");
    fgets(name, sizeof(name), stdin);

    name[strcspn(name, "\n")] = '\0';

    len = strlen(name);

    if (len > 0 && name[0] != ' ')
        printf("%c.", name[0]);

    for (i = 1; i < len; i++)
    {
        if (name[i - 1] == ' ' && name[i] != ' ')
        {
            printf("%c.", name[i]);
        }
    }

    printf("\n");
}
```

Execution

```
Last login: Mon Oct 12 10:58:31 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day44.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day44.c -o day44
[bhavya@Bhavyas-MacBook-Air ~ % ./day44

----- MENU -----
1. Reverse each word in a sentence
2. Print initials of a name
3. Exit
Enter your choice: 1
Enter a sentence: what is special?
Sentence after reversing each word: tahw si ?laiceps

----- MENU -----
1. Reverse each word in a sentence
2. Print initials of a name
3. Exit
Enter your choice: 2
Enter the full name: bhavya rana
b.r.

----- MENU -----
1. Reverse each word in a sentence
2. Print initials of a name
3. Exit
Enter your choice: 3
Exiting program...
bhavya@Bhavyas-MacBook-Air ~ %
```